Cover Page

Hack
for Social Cause

- **Problem Statement/Concept Title**

  Enterprise Cryptographic Discovery & Analysis Tool (ECDAT)

- **Theme**

  Blockchain & Cybersecurity

- **Team Name (Registered on portal)**

  Budding Coders

- **Team ID (Registered on portal)**

  [Add your Team ID]

- **State/UT**

  Tamil Nadu

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Problem - Proposed Solution

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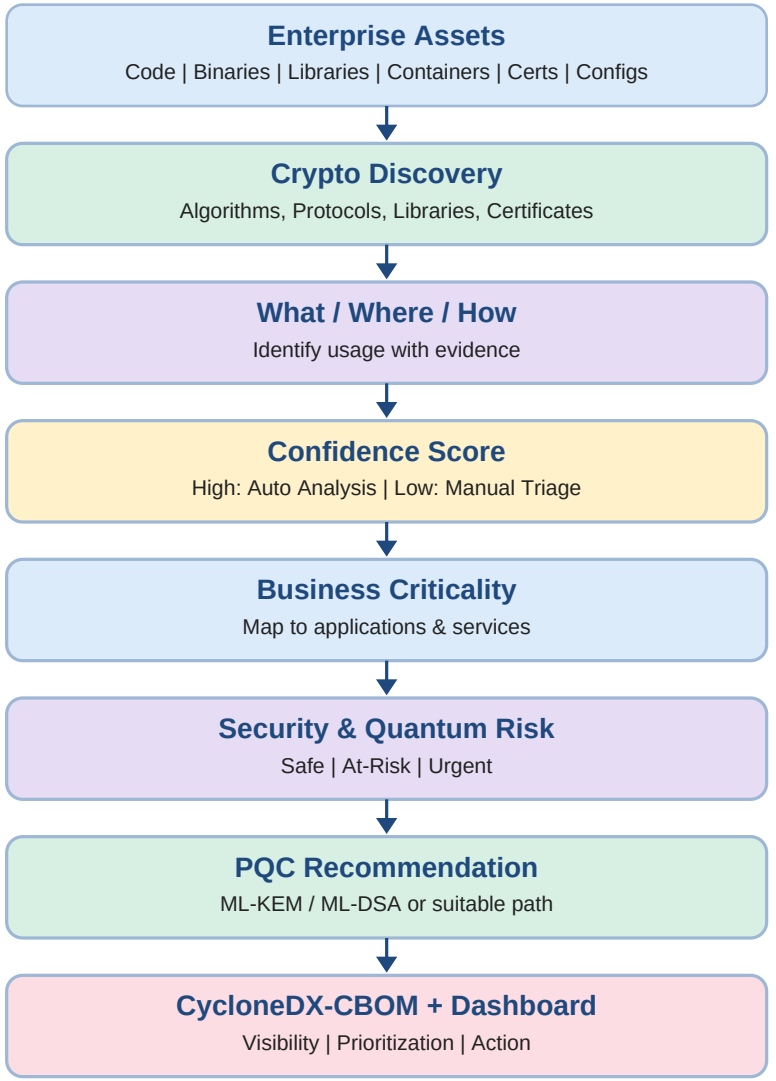
Core problem / Proposed Solution:

- Centralized platform to discover and analyze cryptographic usage across enterprise software environments.
- Scans source code, binaries, libraries, container images, artifacts, configurations, and certificates.
- Identifies cryptographic algorithms, protocols, libraries, and certificates and determines WHAT, WHERE, and HOW they are used.
- Generates cryptographic artifacts with evidence and confidence scores to distinguish reliable findings from uncertain detections.
- Sends low-confidence findings for manual triage instead of treating every detection as valid.
- Correlates cryptographic findings with applications, services, dependencies, and business criticality.
- Provides PQC migration recommendations based on the identified cryptographic use case.
- Generates a CycloneDX-compatible Cryptography Bill of Materials (CBOM) for enterprise-wide cryptographic visibility.



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Technical Approach

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ECDAT - Enterprise Cryptographic Discovery & Analysis Tool

DISCOVER * UNDERSTAND * ASSESS * PRIORITIZE * RECOMMEND

1 Data Ingestion

- Source code, binaries, libraries
- Container images
- Configurations & certificates
- Enterprise software ecosystem

2 Static & Dependency Analysis

- Semgrep + AST: pattern matching, code analysis, dependency scan
- Synt-based SCA: library analysis, version mapping, vulnerable crypto usage

3 Intelligence Layer

- Classification: algorithm/protocol, usage context, lifetime
- Confidence scoring: evidence-based, manual triage for low score
- Correlation: apps, services, business criticality

4 Risk & Quantum Analysis

- Mosca's algorithm, data shelf-life (X + Y vs Z)
- Safe / At-Risk / Urgent
- NIST-approved PQC: ML-KEM / ML-DSA
- Hybrid migration guidance

5 Reporting & Output

- CycloneDX-CBOM (standard, machine-readable)
- Interactive dashboard: inventory, risk, migration
- Export & API access

Technology Stack

Languages	Frameworks	Analysis Tools	Database	Deployment	Standards
Python	FastAPI, React	Semgrep, SCA, LLVM	PostgreSQL	Docker, CI/CD (GitHub Actions)	CycloneDX, PQC (NIST)

Output:

- Complete cryptographic visibility
- Risk-based prioritization
- PQC migration guidance
- CycloneDX-CBOM report
- Integration with CI/CD pipeline

Links & AI Usage:

- GitHub: github.com/ashwinbarat192007-svg
- Demo Video: [Add link]
- Extent of usage of AI tools: [Add details]

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Impact Potential

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Key Benefits:

01 Cryptographic Visibility
Complete view of algorithms, protocols, libraries and certificates

02 Reliable Discovery
Evidence-based confidence across code, binaries, dependencies, configs

03 Risk Prioritization
Algorithm strength, usage context, business criticality

04 Quantum Readiness
Evaluate impact of quantum threats on current crypto assets

05 PQC Migration Guidance
NIST-approved PQC alternatives per use case (RSA/ECC to PQC)

06 Continuous Governance
Up-to-date cryptographic inventory, change monitoring, compliance

Potential Impact on Target Audience / Stakeholders:

Enterprise Security Teams

- Complete centralized visibility of cryptographic usage.
- Faster identification of outdated or risky cryptographic components.
- Reduced manual effort in cryptographic discovery.
- Better prioritization of remediation activities.

Application & DevOps Teams

- Early identification of cryptographic dependencies during development.
- Integration into CI/CD for continuous cryptographic visibility.
- Easier tracking of cryptographic changes across software releases.

Enterprise Management

- Connects technical findings with business-critical applications.
- Enables risk-based prioritization instead of treating every finding equally.
- Supports long-term quantum-readiness planning.

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Feasibility and Risk Analysis

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Feasibility

- Software-only - no specialized hardware required.
- Uses existing enterprise artifacts and security data.
- Modular architecture enables independent development and testing.
- Integrates with CI/CD and security workflows.
- Built on established cryptographic standards and CBOM formats.

Potential Challenges & Risks

- Hidden crypto usage in binaries and dependencies.
- False positives in automated analysis.
- Limited visibility into runtime cryptography.
- Large-scale enterprise environments.
- PQC migration depends on the cryptographic use case.

Strategies to Overcome

- Multi-source evidence correlation.
- Confidence scoring + manual triage.
- Combine code, binary, dependency, configuration & certificate analysis.
- Risk-based prioritization using confidence + business criticality.
- Human-validated PQC recommendations.

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Team

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Team Composition

Team Budding Coders - Sri Sairam Engineering College, Chennai (B.E. CSE - AI & ML)

Roles

- Ashwin Barath E - AI/ML Lead & Backend: detection logic, confidence scoring, risk classification (Python, scikit-learn, TensorFlow), plus SQL/PostgreSQL, CBOM generation and AWS/Docker deployment

Challenges

- Detecting hidden crypto usage inside binaries and dependencies
- Reducing false positives in automated analysis (solved with confidence scoring + manual triage)
- Limited visibility into runtime cryptography
- Choosing the right PQC migration path for each cryptographic use case

Our Journey

- Chose to tackle quantum-era cryptographic risk faced by enterprises
- Designed a modular 5-stage pipeline: ingestion, analysis, intelligence, risk & quantum analysis, CBOM reporting
- Built on open standards: CycloneDX CBOM, NIST PQC (ML-KEM / ML-DSA), Semgrep

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References

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Software Security & Supply Chain

- Software Composition Analysis (SCA)
- Static Application Security Testing (SAST)
- Software Bill of Materials (SBOM)
- Cryptographic Bill of Materials (CBOM)
- CI/CD security integration

Cryptographic & Security Standards

- NIST Post-Quantum Cryptography Standards
- NIST Cybersecurity Framework
- TLS / PKI standards and specifications
- CycloneDX CBOM specification

Post-Quantum Cryptography

- NIST-standardized ML-KEM for post-quantum key establishment
- NIST-standardized ML-DSA for post-quantum digital signatures

Reference Links

- NIST PQC: nist.gov/pqcrypto
- CycloneDX: cyclonedx.org